Hui Li

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Education

The Chinese University of Hong Kong (CUHK), CSE — Sep. 2024 - present
PhD Student, Husky Data Lab | advisor: James Cheng — Hong Kong
- Vector Search, LLM Infrastructure

South China University of Technology (SCUT) — Sep. 2020 - Jul. 2024
B.Eng in Software Engineering | advisor: Prof. Deyou Tang — Guangzhou
- National Scholarship 2020 & National Scholarship 2021;

Contest Experience

ACM International Collegiate Programming Contest (ACM-ICPC)
Asia East Continent Final (2021) **Gold Medal**
+ *6 Regional Gold Medals*: Nanjing; Xi'an; Macau; Jinan (2020 & 2021); Kunming;

China Collegiate Programming Contest (CCPC)
CCPC Final (2021) **Silver Medal First Place**
+ *2 Regional Gold Medals*: Haerbin; Weihai (2020);
+ 1 Regional Silver Medal: Weihai (2021);

GuangDong Province Collegiate Programming Contest (GDCPC 2021), **Second Place**
IOI China Team Selection Competition (CTCS 2019), **Bronze Medals**
National Olympiad in Informatics (NOI 2019), **Bronze Medals**

CCF Big Data and Computing Intelligence Contest (BDCI 2022), **Champion**
Huawei ICT Algorithm Diamond Contest, **Champion**

Publications

SARIA: Persistence-Aware KV Caching with Coordinated DRAM–SSD Reload for LLM Agents
- **Hui Li**, Qihui Zhou, Xiao Yan, Shumeng Zhang, Zijie Dai, Zequn Gong, Mingcong Song, Guoping Long, Hongjie Si, James Cheng
- Submitted

Links: [Paper] [Code]

DRIP: Cracking the Proximity Graph Index for Vector Search
- **Hui Li**, Shiyuan Deng, Xiao Yan, Shangqi Lu, James Cheng
- Proceedings of VLDB' 27

Links: [Paper] [Code]

SAQ: Pushing the Limits of Vector Quantization through CodeAdjustment and Dimension Segmentation
- **Hui Li**, Shiyuan Deng, Xiao Yan, Xiangyu Zhi, James Cheng
- Proceedings of SIGMOD' 26

Links: [Paper] [Code]

DistVS: Large-scale Vector Search with Compute-Memory Disaggregation
- Peiqi Yin, Xiao Yan, Shiyuan Deng, **Hui Li**, Yifan Zhu, Xiangyu Zhi, Jingqi Mao, Ran Xu, James Cheng
- Proceedings of NSDI'26

Links: [Paper]

Gorgeous: Data Layout for Disk-Resident High-Dimensional Vector Search
- Peiqi Yin, Xiao Yan, Qihui Zhou, **Hui Li**, Xiaolu Li, et al.
- Proceedings of SIGMOD' 26

Links: [Paper] [Code]

Internship

Huawei Technologies — Shanghai, July 2026 - Dec. 2026
LLM Infrastructure — *vLLM, Mooncake*
- Explored the design space of DRAM–SSD tiered KV-cache management for efficient LLM inference.

JQ Investments — Shanghai, Jun. 2023 - Sep. 2023
Development Intern — *C++, Quantitative Trading*

Training in quantitative trading domain knowledge and frameworks, including: Quant Research/Market Study; Exchange Order Book Reconstruction; Backtesting Frameworks;

Huawei Technologies Dongguan, Dec. 2022 - Feb. 2023
Software Algorithm Engineer *C++ STL, Algorithm Optimize*

- Optimize performance in the reset scenario and minimize overall computational workload.
- Implement compilation-time conflict detection for lock-free Data Center.

Honor and Scholarship

2026 Huawei Excellent Intern
2025 CUHK The Best Teaching Assistant
2023 Huawei Euler&Gauss Open Source Contribution Scholarship
2022 National Scholarship
2022 SCUT-Tencent Scholarship
2022 SCUT-Huawei Scholarship
2021 National Scholarship
2021 SCUT Pacemaker to Merit Student, Nomination Award (top 0.1%)
2021 Excellent Student Cadre

Teaching

I have served as a teaching assistant for the following courses at CUHK:

- CSCI 2740: Competitive Programming (Fall 2024, Fall 2025)
- CSCI 2520: Data Structures and Applications (Spring 2025)
- CSCI 2100: Data Structures (Spring 2026)

Service

SCUT ACM Training Team May. 2022 - Sep. 2023
vice-captain

- Organize personnel and arrange the tournament;
- Management and operate the training team;

SCUT SE Volunteer Service Team Computer Clinic Sep. 2021 - Sep. 2022
president

- Conducted freshman interview and orientation;
- Managing the operation of the clinic and coordinating with other departments;

GDCCF Youth Programming Ability Improvement Summer Camp Jul. 2021 - Aug. 2021
Instructor

- Course planning and arrangement;
- Course teaching, topic explanation, and comment;

Update at: October 23, 2026